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Appendix: Benchmark Results for Section 3.6.3: Main Program

Table 4a. TOA results. Columns left to right: solar zenith angle, SZA, relative azimuth, AZA, cosine of the view zenith angle (negative for ascending radiation), “exact” results generated with IPOL for Rayleigh scattering, “exact” results generated with IPOL for Aerosol scattering over reflecting surface. One can reproduce this results using our open-source Python script gsit.py.

See Table4a_TOA.xls

Table 4b. Same as Table 4a, except for BOA radiation (positive view cosines).

See Table4b_BOA.xls

Table 4a

Click here to access/download;Table;Table4a_TOA.xlsx

# SZA	AZA	mu	Rayleigh	Aerosol+Surface
45	0	-0.2	1.31857E-01	1.87775E-01
45	0	-0.3	1.19717E-01	1.65770E-01
45	0	-0.4	1.07597E-01	1.43919E-01
45	0	-0.5	9.66534E-02	1.24914E-01
45	0	-0.6	8.73238E-02	1.09385E-01
45	0	-0.7	7.97436E-02	9.70957E-02
45	0	-0.8	7.40228E-02	8.75384E-02
45	0	-0.9	7.05627E-02	8.01726E-02
45	45	-0.2	1.22785E-01	1.37432E-01
45	45	-0.3	1.13270E-01	1.29299E-01
45	45	-0.4	1.03404E-01	1.18425E-01
45	45	-0.5	9.43558E-02	1.07659E-01
45	45	-0.6	8.65802E-02	9.81360E-02
45	45	-0.7	8.02184E-02	9.01544E-02
45	45	-0.8	7.53500E-02	8.36614E-02
45	45	-0.9	7.22449E-02	7.84738E-02
45	90	-0.2	1.17632E-01	8.90322E-02
45	90	-0.3	1.11960E-01	9.04951E-02
45	90	-0.4	1.05141E-01	8.90036E-02
45	90	-0.5	9.84058E-02	8.63224E-02
45	90	-0.6	9.22541E-02	8.33561E-02
45	90	-0.7	8.68261E-02	8.05225E-02
45	90	-0.8	8.21122E-02	7.79903E-02
45	90	-0.9	7.80481E-02	7.58029E-02
45	135	-0.2	1.36168E-01	7.50742E-02
45	135	-0.3	1.30806E-01	7.86107E-02
45	135	-0.4	1.23649E-01	7.95813E-02
45	135	-0.5	1.16028E-01	7.91781E-02
45	135	-0.6	1.08491E-01	7.80961E-02
45	135	-0.7	1.01158E-01	7.67484E-02
45	135	-0.8	9.39060E-02	7.54002E-02
45	135	-0.9	8.63151E-02	7.42491E-02
45	180	-0.2	1.50785E-01	7.33228E-02
45	180	-0.3	1.44516E-01	7.65379E-02
45	180	-0.4	1.36227E-01	7.74871E-02
45	180	-0.5	1.27303E-01	7.79067E-02
45	180	-0.6	1.18310E-01	7.90677E-02
45	180	-0.7	1.09356E-01	7.97093E-02
45	180	-0.8	1.00265E-01	7.65957E-02
45	180	-0.9	9.04610E-02	7.36062E-02

Table 4b

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# SZA	AZA	mu	Rayleigh	Aerosol+Surface
45	0	0.2	9.01739E-02	2.98164E-01
45	0	0.3	9.71640E-02	3.49971E-01
45	0	0.4	9.90347E-02	4.08993E-01
45	0	0.5	9.76689E-02	4.85050E-01
45	0	0.6	9.43856E-02	5.74586E-01
45	0	0.7	8.98800E-02	6.11615E-01
45	0	0.8	8.43994E-02	4.97495E-01
45	0	0.9	7.77215E-02	2.97635E-01
45	45	0.2	8.38213E-02	1.87334E-01
45	45	0.3	8.99192E-02	1.98213E-01
45	45	0.4	9.14588E-02	2.04493E-01
45	45	0.5	9.02424E-02	2.09220E-01
45	45	0.6	8.74911E-02	2.13163E-01
45	45	0.7	8.38433E-02	2.14146E-01
45	45	0.8	7.95429E-02	2.06018E-01
45	45	0.9	7.44621E-02	1.78106E-01
45	90	0.2	7.57570E-02	1.03884E-01
45	90	0.3	7.99524E-02	1.01180E-01
45	90	0.4	8.03044E-02	9.61953E-02
45	90	0.5	7.86288E-02	9.11179E-02
45	90	0.6	7.60858E-02	8.71624E-02
45	90	0.7	7.32872E-02	8.50085E-02
45	90	0.8	7.05331E-02	8.51769E-02
45	90	0.9	6.79616E-02	8.83164E-02
45	135	0.2	7.79772E-02	7.79305E-02
45	135	0.3	8.06257E-02	7.29980E-02
45	135	0.4	7.92412E-02	6.67581E-02
45	135	0.5	7.59467E-02	6.08436E-02
45	135	0.6	7.20904E-02	5.61450E-02
45	135	0.7	6.84131E-02	5.32465E-02
45	135	0.8	6.53628E-02	5.29343E-02
45	135	0.9	6.33962E-02	5.74111E-02
45	180	0.2	8.19092E-02	7.31254E-02
45	180	0.3	8.40211E-02	6.77385E-02
45	180	0.4	8.17565E-02	6.11800E-02
45	180	0.5	7.74517E-02	5.49971E-02
45	180	0.6	7.26058E-02	5.00169E-02
45	180	0.7	6.80585E-02	4.67738E-02
45	180	0.8	6.43457E-02	4.60250E-02
45	180	0.9	6.20719E-02	5.00716E-02